



ACE - LIBRARY BOOK INVENTORY SYSTEM

UNIT – I MINI PROJECT

anveshkumaraavula@outlook.com
[Email address]

//Mini-Project Name: "ACE Library Book Inventory System" -

// a) adding,

// b) viewing, and

// c) checking out of books.

package miniprojects;

import java.util.ArrayList;

import java.util.Scanner;

class Book {

int bookId;

 String title;

 String author;

boolean available;

 Book(**int** id, String title, String author) {

this.bookId = id;

this.title = title;

this.author = author;

this.available = **true**;

 }

void displayBook() {

 System.**out**.println("-----");

 System.**out**.println("Book ID : " + bookId);

 System.**out**.println("Title : " + title);

 System.**out**.println("Author : " + author);

 System.**out**.println("Status : " + (available ? "Available" :
"Checked Out"));

 } }

```
public class ACELibraryInventory {  
  
    static ArrayList<Book> books = new ArrayList<>();  
    static Scanner sc = new Scanner(System.in);  
  
    static void addBook() {  
  
        System.out.print("Enter Book ID : ");  
        int id = sc.nextInt();  
        sc.nextLine();  
  
        System.out.print("Enter Title : ");  
        String title = sc.nextLine();  
  
        System.out.print("Enter Author : ");  
        String author = sc.nextLine();  
  
        books.add(new Book(id, title, author));  
  
        System.out.println("Book Added Successfully!");  
    }  
  
    static void viewBooks() {  
  
        if (books.isEmpty()) {  
            System.out.println("No books available.");  
            return;  
        }  
  
        for (Book b : books)
```

```
        b.displayBook();
    }

    static void checkoutBook() {

        System.out.print("Enter Book ID : ");
        int id = sc.nextInt();

        for (Book b : books) {

            if (b.bookId == id) {

                if (b.available) {

                    b.available = false;
                    System.out.println("Book Checked Out Successfully!");

                } else {

                    System.out.println("Book Already Issued.");
                }

                return;
            }
        }

        System.out.println("Book Not Found.");
    }
```

```
public static void main(String[] args) {  
  
    int choice;  
  
    do {  
  
        System.out.println("\n===== ACE Library Book Inventory  
System =====");  
        System.out.println("1. Add Book");  
        System.out.println("2. View Books");  
        System.out.println("3. Check Out Book");  
        System.out.println("4. Exit");  
  
        System.out.print("Enter Choice : ");  
        choice = sc.nextInt();  
  
        switch (choice) {  
  
            case 1:  
                addBook();  
                break;  
  
            case 2:  
                viewBooks();  
                break;  
  
            case 3:  
                checkoutBook();  
                break;
```

case 4:

System.*out*.println("Thank You!");

break;

default:

System.*out*.println("Invalid Choice");

}

} **while** (choice != 4);

}

}

OUT PUT:-

===== ACE Library Book Inventory System =====

1. Add Book

2. View Books

3. Check Out Book

4. Exit

Enter Choice : 2

No books available.

===== ACE Library Book Inventory System =====

1. Add Book

2. View Books

3. Check Out Book

4. Exit

Enter Choice : 1

Enter Book ID : 101

Enter Title : Java Programming

Enter Author : ASDF

Book Added Successfully!

===== ACE Library Book Inventory System =====

1. Add Book
2. View Books
3. Check Out Book
4. Exit

Enter Choice : 1

Enter Book ID : 103

Enter Title : SOFTWARE ENGINEERING

Enter Author : CPS

Book Added Successfully!

===== ACE Library Book Inventory System =====

1. Add Book
2. View Books
3. Check Out Book
4. Exit

Enter Choice : 2

Book ID : 101

Title : Java Programming

Author : ASDF

Status : Available

Book ID : 103

Title : SOFTWARE ENGINEERING

Author : CPS

Status : Available